SiteGuard Agent -- OpenCV AI Competition 2026 Proposal

Team: SiteGuard Agent (solo -- Solana Minter, "Dy", Green Robin)

Contact: solanaminter@gmail.com | GitHub: github.com/solanaminter

Problem statement & real-world impact

Construction sites injure workers every year from preventable causes -- chiefly missing hardhats and workers entering restricted zones (excavations, crane swing areas, live electrical). Safety officers cannot watch every camera, and fixed alarms cry wolf until crews ignore them. SiteGuard Agent is an agentic vision monitor that reasons about risk, keeps a human in the loop, and revises its decisions when later vision changes the picture -- cutting alarm fatigue while catching the violations that matter.

Planned OpenCV 5 image/video analysis

YOLOX-s (official OpenCV Model Zoo, ONNX, cv2.dnn) person detection at 1280x720/15fps; classical HSV hardhat segmentation with dome-shape analysis on the head ROI; pointPolygonTest restricted-zone checks; Farneback optical flow motion signal; IoU centroid tracking with hardhat majority vote -- orchestrated per frame through imgproc/dnn/video modules.

Planned AWS architecture & services

Docker -> ECR -> ECS Fargate behind an Application Load Balancer; S3 evidence bucket; CloudWatch structured JSON logs; everything declared in CloudFormation and deployed with one script. FastAPI review console for human approve/reject.

Target users

Site safety officers (review console), site managers (escalation alerts + evidence), construction workers (beneficiaries).

Evaluation method & judge demo

8 hand-labeled scenes incl. failure probes (detection P/R, hardhat P/R, zone accuracy, CPU latency); a scripted 27s multi-camera demo proving the agentic loop -- pending approval WITHDRAWN when a hardhat appears (trace f157), restricted-zone violation auto-ESCALATED (trace f228) -- with a timestamped JSON trace as evidence. Judges get the video, the trace, and a live endpoint or screen-share.

Paths

Agentic Vision path (primary target). Not pursuing COOL.

Team bio

Solana Minter ("Dy") -- solo builder at Green Robin. Ships end-to-end: computer vision pipelines, agentic systems, and AWS delivery. Recent hackathon submissions: Permit Pal (Agents for Humans Hackathon) and Churn Radar (AI Builders Hackathon), each with repo + demo video.